

TECHSHIP INTERNSHIP

Day 8 – MongoDB Fundamentals & Database Operations

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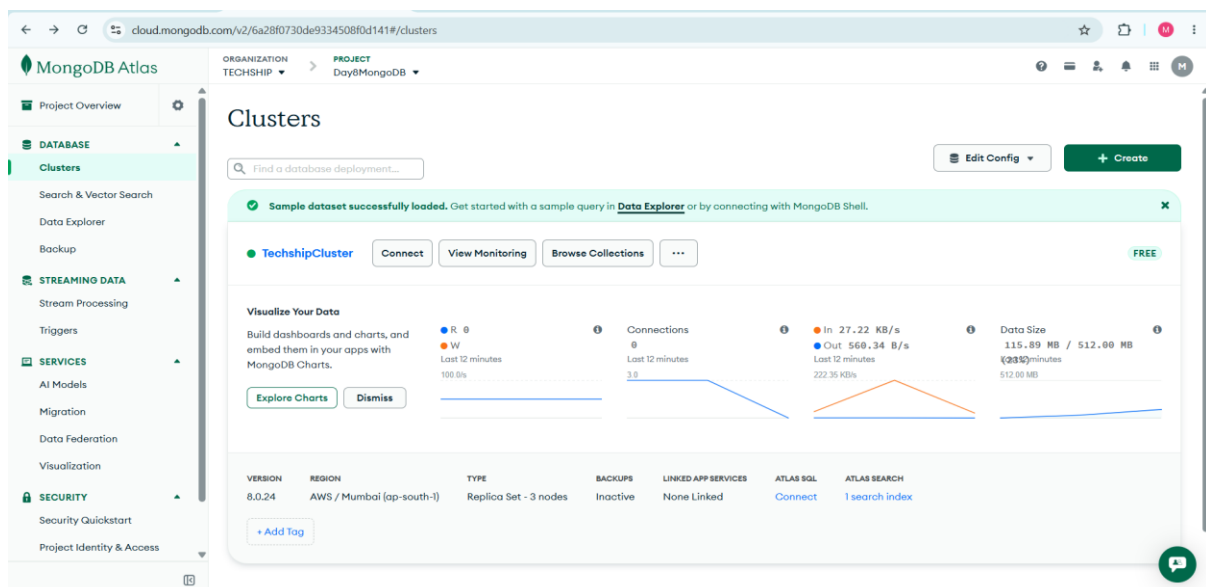
During Day 8 of the TECHSHIP Internship Program, I learned the fundamentals of databases and their importance in storing, organizing, and managing data efficiently. I gained hands-on experience in creating databases, collections, and documents using MongoDB Atlas. I also learned how CRUD operations (Create, Read, Update, and Delete) are used to manage data in real-world applications.

I understood the key differences between SQL and MongoDB. SQL databases store data in tables with rows and columns and use a fixed schema. In contrast, MongoDB is a NoSQL database that stores data in flexible JSON-like documents within collections. This flexibility makes MongoDB suitable for modern web applications that handle different types of data.

While using MongoDB Atlas, I faced some challenges in navigating the interface, creating clusters, and locating options for creating databases and collections. However, with practice and exploration, I became more comfortable using Atlas and gained a better understanding of how databases support applications such as student management systems and task management systems.

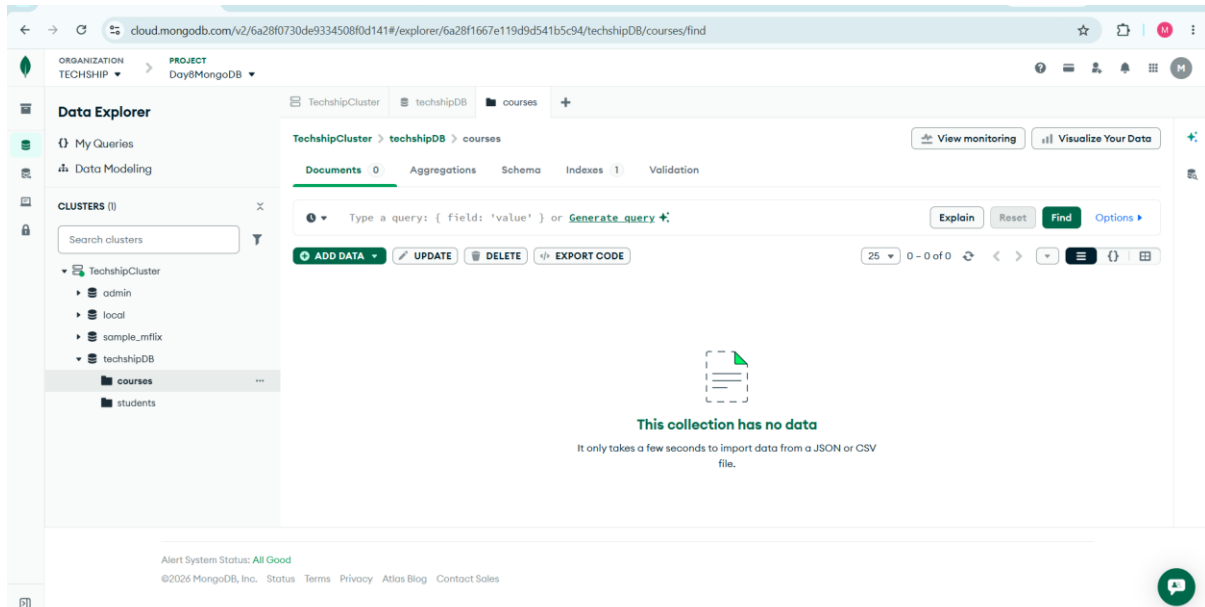
PRACTICAL EXERCISES

Activity 1 – Create a MongoDB Atlas Account



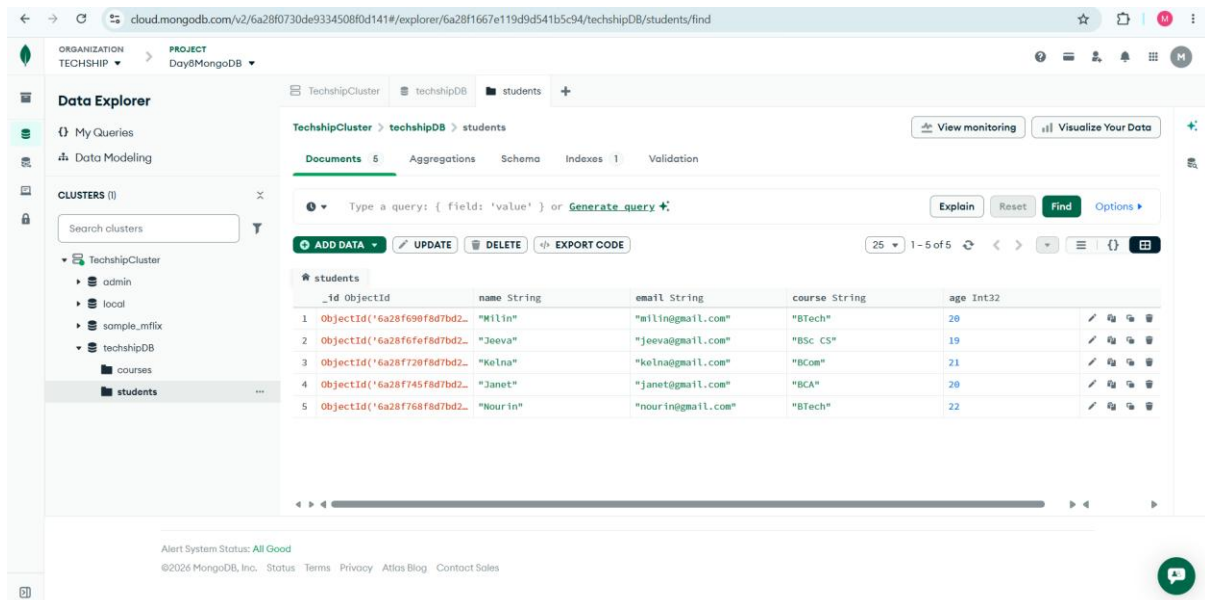
A MongoDB Atlas account was created successfully. A new project and free cluster were set up using MongoDB Atlas. The cluster dashboard was accessed and verified.

Activity 2 – Create a Database



A database named techshipDB was created. Two collections, students and courses, were created within the database to store student and course information.

Activity 3 – Insert Student Records



Five student documents were inserted into the students collection. Each document contained fields such as name, email, course, and age. The records were successfully stored and verified.

Activity 4 – Read Data

The screenshot shows the MongoDB Data Explorer interface. The left sidebar displays the 'clusters' list with 'TechshipCluster' expanded, showing 'students' as the selected collection. The main panel shows the 'students' collection with 5 documents. The first document is expanded, showing its JSON structure:

```
{
  "_id": {},
  "name": "Milin",
  "email": "milin@gmail.com",
  "course": "BTech",
  "age": 20
}
```

The interface includes a search bar, a query input field, and buttons for 'ADD DATA', 'UPDATE', 'DELETE', and 'EXPORT CODE'. The bottom status bar indicates 'Alert System Status: All Good' and '©2026 MongoDB, Inc.'.

The screenshot shows the MongoDB Data Explorer interface. The left sidebar displays the 'clusters' list with 'TechshipCluster' expanded, showing 'students' as the selected collection. The main panel shows the 'students' collection with 5 documents. The first document is expanded, showing its JSON structure:

```
{
  "_id": {},
  "name": "Kelna",
  "email": "kelna@gmail.com",
  "course": "BCom",
  "age": 21
}
```

The interface includes a search bar, a query input field, and buttons for 'ADD DATA', 'UPDATE', 'DELETE', and 'EXPORT CODE'. The bottom status bar indicates 'Alert System Status: All Good' and '©2026 MongoDB, Inc.'.

All student records were retrieved and viewed from the students collection. The document structure, fields, values, and collection layout were observed. MongoDB automatically generated a unique `_id` field for each document.

Activity 5 – Update Records

The screenshot shows the MongoDB Cloud Portal interface. On the left, the 'Data Explorer' sidebar shows the 'students' collection under the 'techshipDB' database. The main panel displays the 'students' collection with a list of documents. The 'UPDATE' button is highlighted, indicating that a record is being updated. The documents shown are:

```
{ "_id": "1", "name": "Milin", "email": "milin@gmail.com", "course": "BCA", "age": 21 }
{ "_id": "2", "name": "Jeeva", "email": "jeeva@gmail.com", "course": "BSc CS", "age": 19 }
{ "_id": "3", "name": "Kelna", "email": "kelna@gmail.com", "course": "BCA", "age": 21 }
```

An existing student record was updated by modifying the course and age fields. The updated information was successfully saved and verified in the collection.

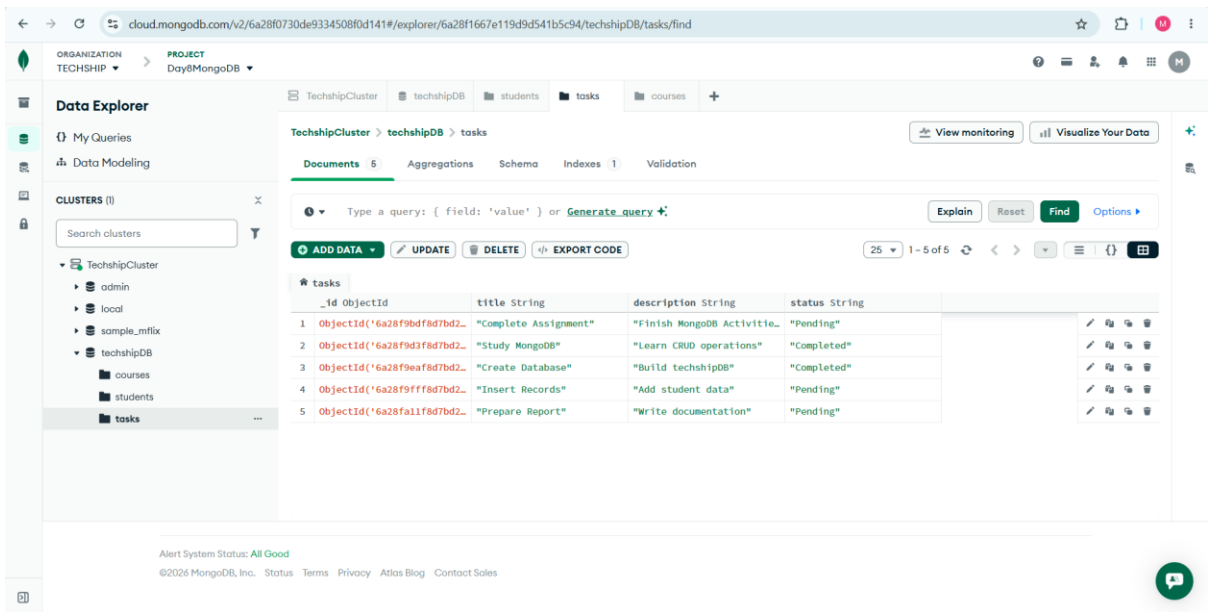
Activity 6 – Delete Records

The screenshot shows the MongoDB Cloud Portal interface. On the left, the 'Data Explorer' sidebar shows the 'students' collection under the 'techshipDB' database. The main panel displays the 'students' collection with a list of documents. The 'DELETE' button is highlighted, indicating that a record is being deleted. The documents shown are:

	_id ObjectId	name String	email String	course String	age Int32
1	ObjectId('6a28f698f8d7bd2...')	Milina	milina@gmail.com	BCA	21
2	ObjectId('6a28f6f6f8d7bd2...')	Jeeva	jeeva@gmail.com	BSc CS	19
3	ObjectId('6a28f728f8d7bd2...')	Kelna	kelna@gmail.com	BCom	21
4	ObjectId('6a28f745f8d7bd2...')	Janet	janet@gmail.com	BCA	20

One student document was deleted from the students collection. The collection was checked afterward to confirm that the record had been removed successfully.

Activity 7 – Design a Collection



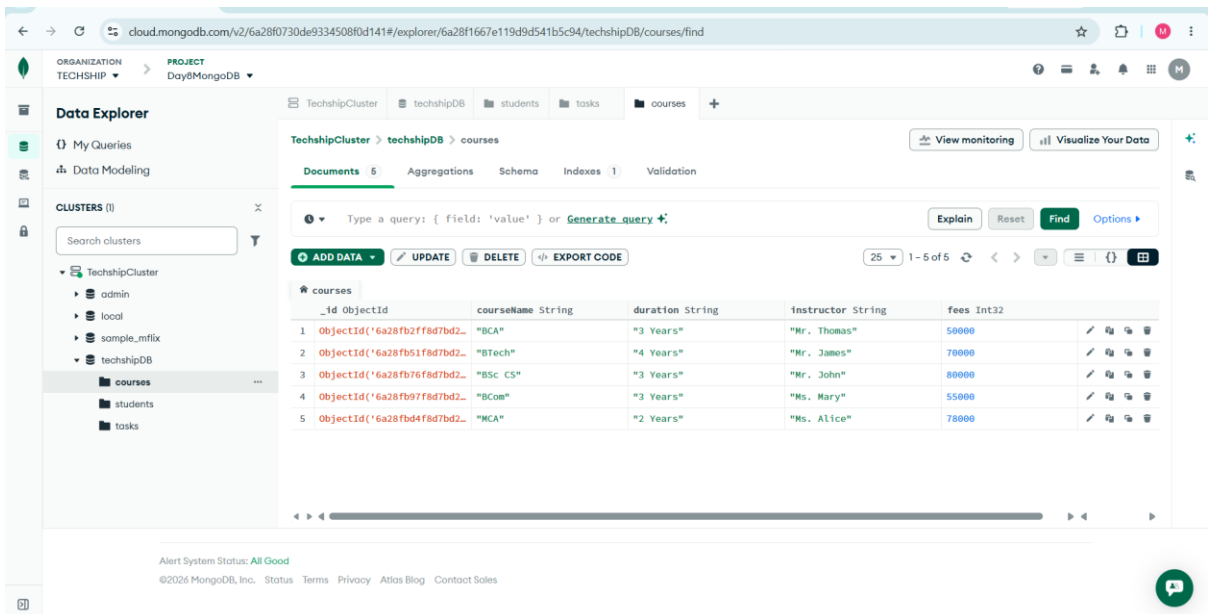
The screenshot shows the MongoDB Data Explorer interface. The left sidebar displays the cluster structure: TechshipCluster > techshipDB > tasks. The main panel shows the 'tasks' collection with 5 documents. The documents are listed in a table with columns: _id, title, description, and status.

_id	title	description	status
ObjectId('6a28f9bdf8d7bd2...')	"Complete Assignment"	"Finish MongoDB Activitie.."	"Pending"
ObjectId('6a28f9d3f8d7bd2...')	"Study MongoDB"	"Learn CRUD operations"	"Completed"
ObjectId('6a28f9eaf8d7bd2...')	"Create Database"	"Build techshipDB"	"Completed"
ObjectId('6a28f9fff8d7bd2...')	"Insert Records"	"Add student data"	"Pending"
ObjectId('6a28fa1f8d7bd2...')	"Prepare Report"	"Write documentation"	"Pending"

A collection named tasks was created for a Task Management System. Five sample task documents containing title, description, and status fields were inserted successfully.

CHALLENGE ACTIVITIES

Challenge 1 – Course Management Collection



The screenshot shows the MongoDB Data Explorer interface. The left sidebar displays the cluster structure: TechshipCluster > techshipDB > courses. The main panel shows the 'courses' collection with 5 documents. The documents are listed in a table with columns: _id, courseName, duration, instructor, and fees.

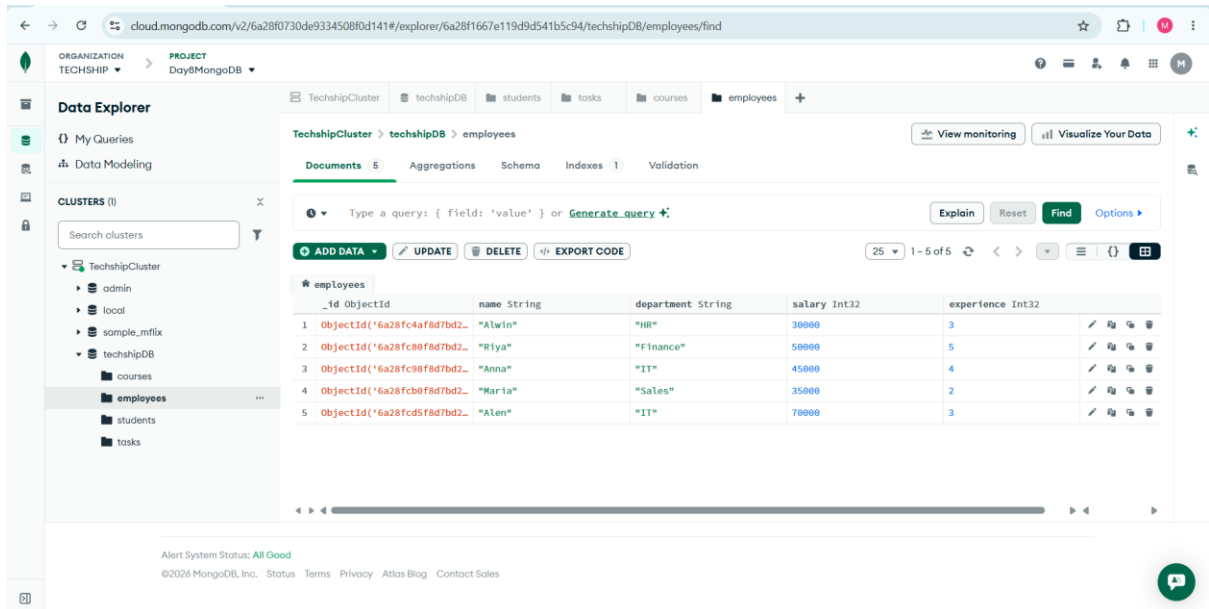
_id	courseName	duration	instructor	fees
ObjectId('6a28fb2ff8d7bd2...')	"BCA"	"3 Years"	"Mr. Thomas"	50000
ObjectId('6a28fb51f8d7bd2...')	"BTech"	"4 Years"	"Mr. James"	70000
ObjectId('6a28fb76f8d7bd2...')	"BSc CS"	"3 Years"	"Mr. John"	80000
ObjectId('6a28fb97f8d7bd2...')	"BCom"	"3 Years"	"Ms. Mary"	55000
ObjectId('6a28fb4f8d7bd2...')	"MCA"	"2 Years"	"Ms. Alice"	78000

The courses collection was populated with five course records. Each document contained course name, duration, instructor, and fees information. The records were stored and verified successfully.

Challenge 2 – Employee Database

An employees collection was created with employee details such as name, department, salary, and experience.

Create Operation

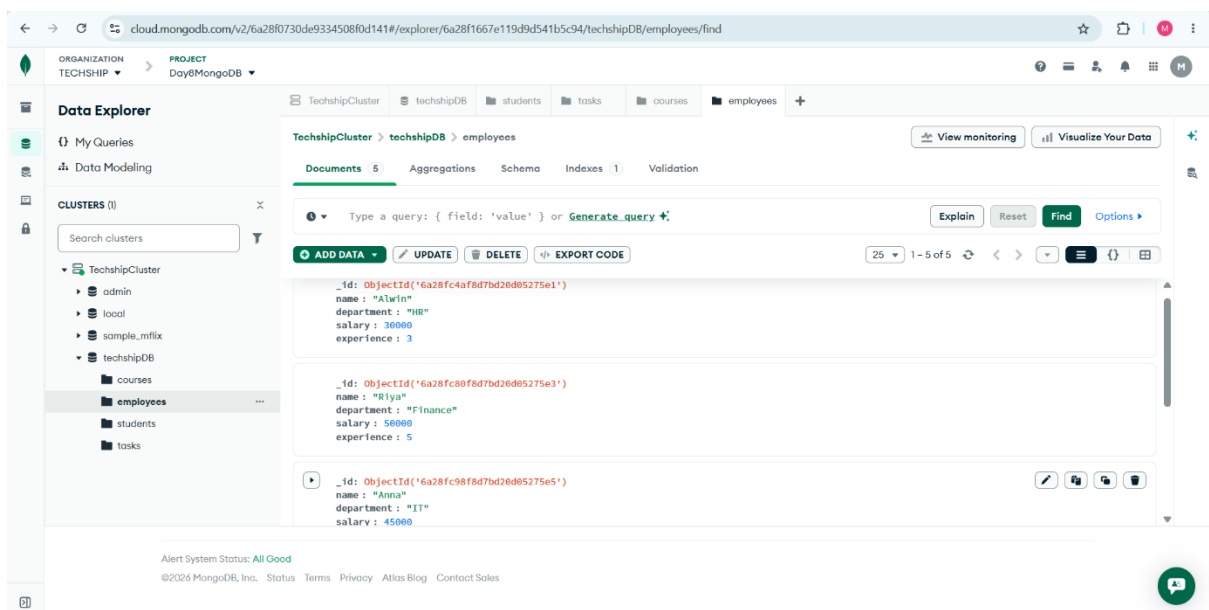


The screenshot shows the MongoDB Cloud Portal interface. The left sidebar displays the 'Data Explorer' with a tree view of clusters. The main panel shows the 'employees' collection in the 'techshipDB' database. The 'Documents' tab is active, displaying a table with 5 documents. The table has columns for _id, name, department, salary, and experience. The documents are as follows:

	_id	name	department	salary	experience
1	ObjectId('6a28fc4af8d7bd2...')	"Alwin"	"HR"	30000	3
2	ObjectId('6a28fc89f8d7bd2...')	"Riya"	"Finance"	50000	5
3	ObjectId('6a28fc98f8d7bd2...')	"Anna"	"IT"	45000	4
4	ObjectId('6a28fcb0f8d7bd2...')	"Maria"	"Sales"	35000	2
5	ObjectId('6a28fcd5f8d7bd2...')	"Aleen"	"IT"	70000	3

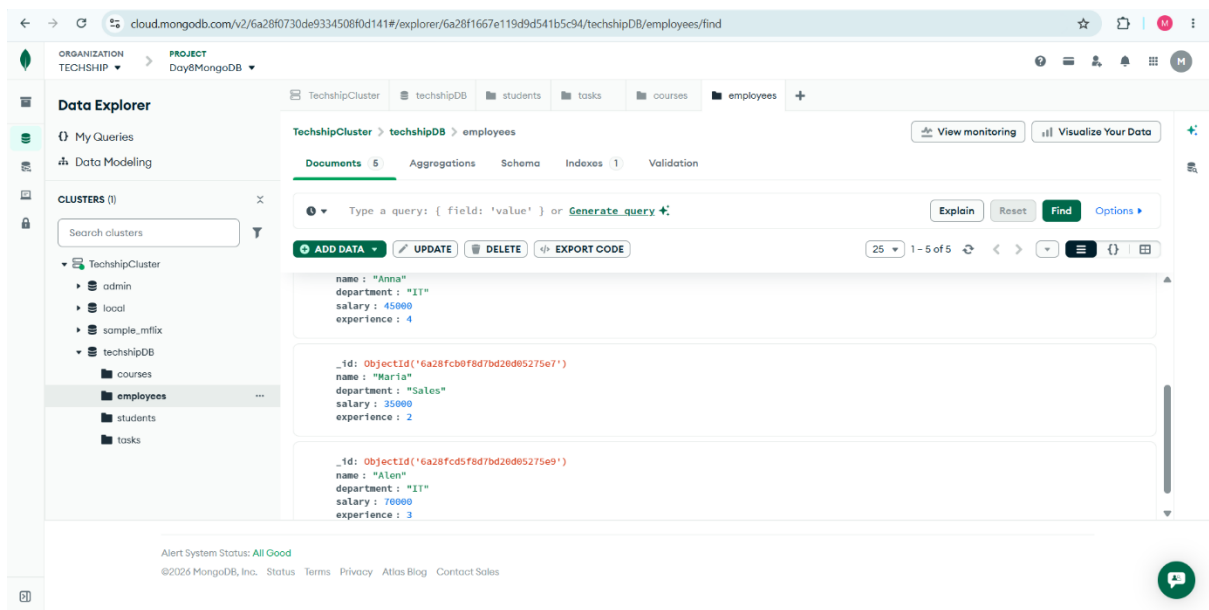
Employee records were inserted into the employees collection. Each document contained details such as name, department, salary, and experience.

Read Operation



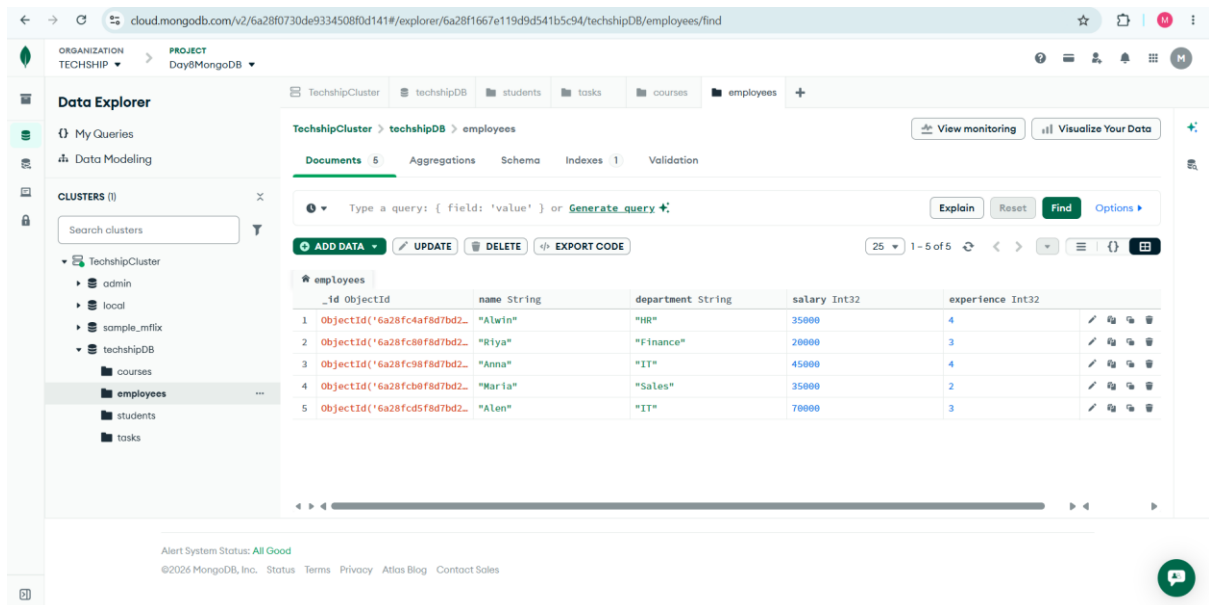
The screenshot shows the MongoDB Cloud Portal interface. The left sidebar displays the 'Data Explorer' with a tree view of clusters. The main panel shows the 'employees' collection in the 'techshipDB' database. The 'Documents' tab is active, displaying a list of 3 documents. The documents are as follows:

```
{ "_id": "ObjectId('6a28fc4af8d7bd26d85275e1')",  
  "name": "Alwin",  
  "department": "HR",  
  "salary": 30000,  
  "experience": 3 }  
  
{ "_id": "ObjectId('6a28fc89f8d7bd26d85275e3')",  
  "name": "Riya",  
  "department": "Finance",  
  "salary": 50000,  
  "experience": 5 }  
  
{ "_id": "ObjectId('6a28fc98f8d7bd26d85275e5')",  
  "name": "Anna",  
  "department": "IT",  
  "salary": 45000 }
```



All employee records were retrieved and viewed from the employees collection. The stored data and document structure were verified successfully.

Update Operation



An existing employee record was updated by modifying fields such as salary and experience. The updated values were saved and verified in the collection.

Delete Operation

The screenshot shows the MongoDB Cloud Portal interface. On the left, the 'Data Explorer' sidebar lists clusters and databases. The 'employees' collection is selected. The main panel shows a table of employee records. The first record, with name 'Alwin', is highlighted. Below the table, there are buttons for 'ADD DATA', 'UPDATE', 'DELETE', and 'EXPORT CODE'. The 'DELETE' button is highlighted, indicating the deletion operation.

_id	ObjectID	name	String	department	String	salary	Int32	experience	Int32
1	ObjectId('6a28fc4af8dbd2...')	Alwin		HR		35000		4	
2	ObjectId('6a28fc98f8dbd2...')	Riya		Finance		28000		3	
3	ObjectId('6a28fc98f8dbd2...')	Anna		IT		45000		4	
4	ObjectId('6a28fc98f8dbd2...')	Maria		Sales		35000		2	

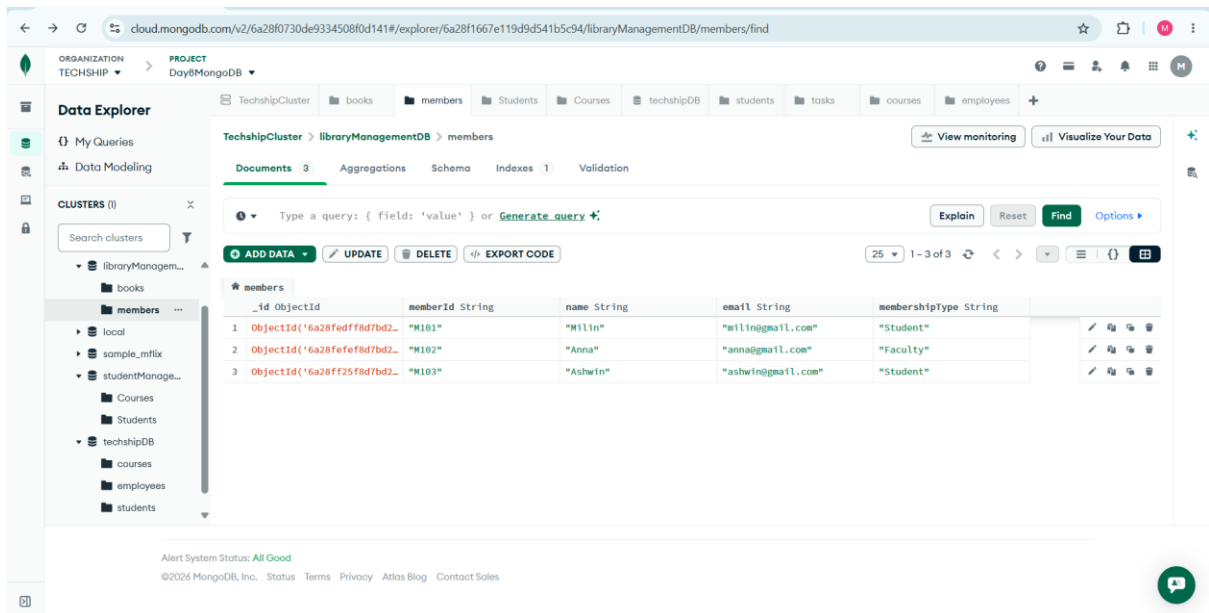
One employee record was deleted from the employees collection. The collection was checked afterward to confirm that the document had been removed successfully.

Challenge 3 – Design Your Own Database

Library Management Database

The screenshot shows the MongoDB Cloud Portal interface for the 'Library Management Database'. The 'books' collection is selected. The main panel shows a table of book records. The first record, with title 'Introduction to Java', is highlighted. Below the table, there are buttons for 'ADD DATA', 'UPDATE', 'DELETE', and 'EXPORT CODE'. The 'DELETE' button is highlighted, indicating the deletion operation.

_id	ObjectID	bookId	String	title	String	author	String	category	String	availabl
1	ObjectId('6a28feb8f8dbd2...')	B101		Introduction to Java		Herbert Schildt		Programming		true
2	ObjectId('6a28feb8f8dbd2...')	B102		Database Systems		Elmasri		Database		true
3	ObjectId('6a28feb8f8dbd2...')	B103		Computer Networks		Andrew Tanenbaum		Networking		false



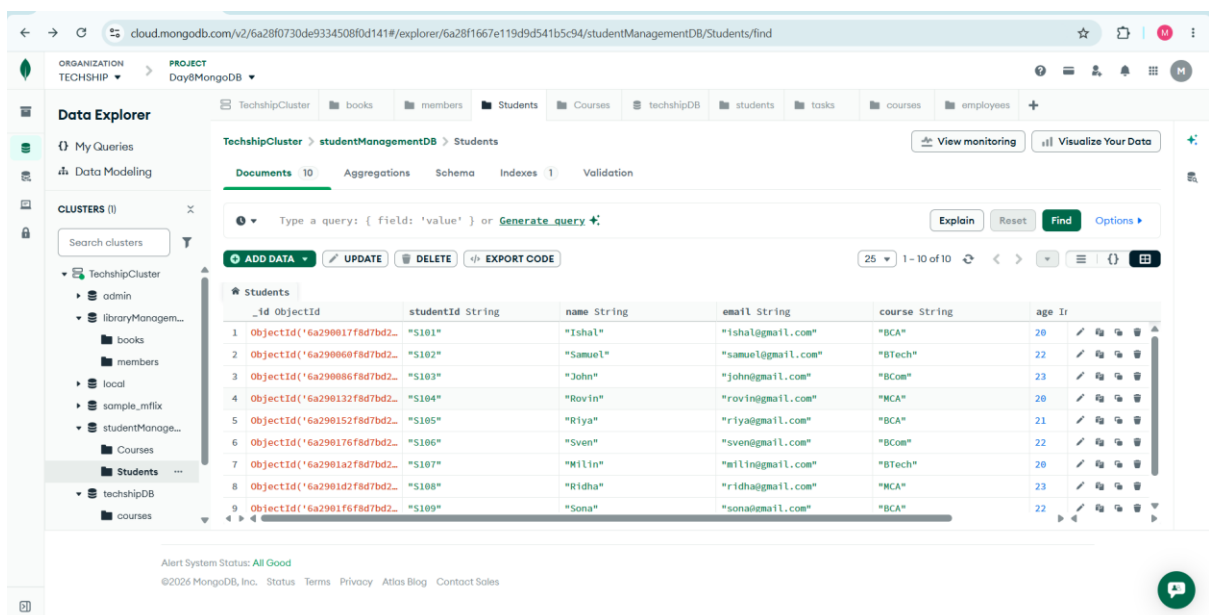
A database named libraryManagementDB was created. Two collections, books and members, were designed and populated with sample documents. This activity demonstrated how MongoDB can be used to model a real-world library management system.

ASSIGNMENT

Student Management Database

A database named studentManagementDB was created successfully in MongoDB Atlas to store student and course information. Two collections, students and courses, were created within the studentManagementDB database. These collections were used to organize student and course records separately.

Create Operation



cloud.mongodb.com/v2/6a28f0730de9334508f0d141#/explorer/6a28f1667e119d9d541b5c94/studentManagementDB/Students/find

ORGANIZATION: TECHSHIP | PROJECT: Day8MongoDB

Clusters: TechshipCluster

Collection: studentManagementDB > Students

Documents: 10 | Aggregations: 0 | Schema: 1 | Indexes: 1 | Validation: 0

Type a query: { field: 'value' } or [Generate query](#)

Buttons: ADD DATA, UPDATE, DELETE, EXPORT CODE

Display: 25 | 1 - 10 of 10

_id	studentId	name	email	course	age
ObjectID('6a29060f8d7bd2...')	"S102"	"Samuel"	"samuel@gmail.com"	"BTech"	22
ObjectID('6a29060f8d7bd2...')	"S103"	"John"	"john@gmail.com"	"BCom"	23
ObjectID('6a290132f8d7bd2...')	"S104"	"Rovin"	"rovin@gmail.com"	"MCA"	20
ObjectID('6a290152f8d7bd2...')	"S105"	"Riya"	"riya@gmail.com"	"BCA"	21
ObjectID('6a290176f8d7bd2...')	"S106"	"Sven"	"sven@gmail.com"	"BCom"	22
ObjectID('6a2901a2f8d7bd2...')	"S107"	"Milin"	"milin@gmail.com"	"BTech"	20
ObjectID('6a2901d2f8d7bd2...')	"S108"	"Ridha"	"ridha@gmail.com"	"MCA"	23
ObjectID('6a2901f6f8d7bd2...')	"S109"	"Sona"	"sona@gmail.com"	"BCA"	22
ObjectID('6a290226f8d7bd2...')	"S110"	"Sam"	"sam@gmail.com"	"BTech"	20

Alert System Status: All Good

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Ten student documents were inserted into the students collection. Each document contained fields such as Student ID, Name, Email, Course, and Age.

cloud.mongodb.com/v2/6a28f0730de9334508f0d141#/explorer/6a28f1667e119d9d541b5c94/studentManagementDB/Courses/find

ORGANIZATION: TECHSHIP | PROJECT: Day8MongoDB

Clusters: TechshipCluster

Collection: studentManagementDB > Courses

Documents: 5 | Aggregations: 0 | Schema: 1 | Indexes: 1 | Validation: 0

Type a query: { field: 'value' } or [Generate query](#)

Buttons: ADD DATA, UPDATE, DELETE, EXPORT CODE

Display: 25 | 1 - 5 of 5

_id	courseId	courseName	duration	instructor
ObjectID('6a2903a4f8d7bd2...')	"C101"	"BCA"	"3 Years"	"Mr. Thomas"
ObjectID('6a2903c4f8d7bd2...')	"C102"	"BTech"	"4 Years"	"Mr. James"
ObjectID('6a2903dcf8d7bd2...')	"C103"	"BCom"	"3 Years"	"Mr. John"
ObjectID('6a290405f8d7bd2...')	"C104"	"BSc CS"	"3 Years"	"Ms. Alice"
ObjectID('6a29042af8d7bd2...')	"C105"	"MCA"	"2 Years"	"Ms. Anna"

Alert System Status: All Good

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Five course documents were inserted into the courses collection. Each document contained Course ID, Course Name, Duration, and Instructor details.

Read Operation

The screenshot shows the MongoDB Cloud Explorer interface. The left sidebar displays the database structure, with 'Students' selected under 'techshipDB'. The main panel shows the 'Students' collection in the 'studentManagementDB' database. The 'Documents' tab is active, showing a list of documents. The first document is:

```
{ "_id": "ObjectId('6a298017f8d7bd28d05275ff')", "studentId": "5101", "name": "Isha", "email": "isha@gmail.com", "course": "BCA", "age": 20 }
```

The second document is:

```
{ "_id": "ObjectId('6a298060f8d7bd28d0527681')", "studentId": "5102", "name": "Samuel", "email": "samuel@gmail.com", "course": "BTech", "age": 22 }
```

The third document is:

```
{ "_id": "ObjectId('6a298132f8d7bd28d0527686')", "studentId": "5103", "name": "Thomas", "email": "thomas@gmail.com", "course": "BCA", "age": 20 }
```

The screenshot shows the MongoDB Cloud Explorer interface. The left sidebar displays the database structure, with 'Courses' selected under 'techshipDB'. The main panel shows the 'Courses' collection in the 'studentManagementDB' database. The 'Documents' tab is active, showing a list of documents. The first document is:

```
{ "_id": "ObjectId('6a2983a4f8d7bd28d052761a')", "courseId": "C101", "courseName": "BTech", "duration": "4 Years", "instructor": "Mr. Thomas" }
```

The second document is:

```
{ "_id": "ObjectId('6a2983c4f8d7bd28d052761c')", "courseId": "C102", "courseName": "BTech", "duration": "4 Years", "instructor": "Mr. James" }
```

The third document is:

```
{ "_id": "ObjectId('6a2983dcf8d7bd28d052761e')", "courseId": "C103", "courseName": "BCA", "duration": "4 Years", "instructor": "Mr. Thomas" }
```

All records from the students and courses collections were retrieved and viewed successfully. The stored data and document structure were verified.

Update Operation

The screenshot shows the MongoDB Cloud Portal interface. The left sidebar displays the 'Data Explorer' with a tree view of the database structure. The main panel shows the 'Courses' collection in the 'studentManagementDB' database. A query is entered in the search bar: `{ field: 'value' }` or `Generate query`. Below the search bar, there are buttons for 'ADD DATA', 'UPDATE', 'DELETE', and 'EXPORT CODE'. The 'UPDATE' button is highlighted. The table below shows the updated course record.

	_id	ObjectId	courseId	String	courseName	String	duration	String	instructor	String
1	ObjectId('6a2903a4f8d7bd2...')	"C101"	"BTech"	"4 Years"	"Mr. Thomas"					
2	ObjectId('6a2903c4f8d7bd2...')	"C102"	"BTech"	"4 Years"	"Mr. James"					
3	ObjectId('6a2903dcf8d7bd2...')	"C103"	"BCom"	"3 Years"	"Mr. John"					
4	ObjectId('6a290405f8d7bd2...')	"C104"	"BSc CS"	"3 Years"	"Ms. Alice"					
5	ObjectId('6a29042af8d7bd2...')	"C105"	"MCA"	"2 Years"	"Ms. Anna"					

An existing courses record was updated by changing the course name and duration fields. The updated information was successfully reflected in the collection.

The screenshot shows the MongoDB Cloud Portal interface. The left sidebar displays the 'Data Explorer' with a tree view of the database structure. The main panel shows the 'Students' collection in the 'studentManagementDB' database. A query is entered in the search bar: `{ field: 'value' }` or `Generate query`. Below the search bar, there are buttons for 'ADD DATA', 'UPDATE', 'DELETE', and 'EXPORT CODE'. The 'UPDATE' button is highlighted. The table below shows the updated student record.

	_id	ObjectId	studentId	String	name	String	email	String	course	String	age	Int
1	ObjectId('6a290017f8d7bd2...')	"S101"	"Ishal"	"ishal@gmail.com"	"BCA"	20						
2	ObjectId('6a290060f8d7bd2...')	"S102"	"Samuel"	"samuel@gmail.com"	"BTech"	24						
3	ObjectId('6a290086f8d7bd2...')	"S103"	"John"	"john@gmail.com"	"BTech"	22						
4	ObjectId('6a290132f8d7bd2...')	"S104"	"Rovin"	"rovin@gmail.com"	"MCA"	20						
5	ObjectId('6a290152f8d7bd2...')	"S105"	"Riya"	"riya@gmail.com"	"BCA"	21						
6	ObjectId('6a290176f8d7bd2...')	"S106"	"Sven"	"sven@gmail.com"	"BCom"	22						
7	ObjectId('6a2901a2f8d7bd2...')	"S107"	"Milin"	"milin@gmail.com"	"BTech"	20						
8	ObjectId('6a2901d2f8d7bd2...')	"S108"	"Ridha"	"ridha@gmail.com"	"MCA"	23						
9	ObjectId('6a2901f6f8d7bd2...')	"S109"	"Sona"	"sona@gmail.com"	"BCA"	22						

An existing student record was updated by changing the course and age fields. The updated information was successfully reflected in the collection.

Delete Operation

The screenshot displays the MongoDB Cloud Portal interface. The top navigation bar shows the organization 'TECHSHIP' and the project 'Day8MongoDB'. The left sidebar contains the 'Data Explorer' section with a search bar and a tree view of clusters. The main panel shows the 'Students' collection in the 'studentManagementDB' database. A query editor is active with the query: `{ field: 'value' }` or `Generate query`. Below the query editor, there are buttons for 'ADD DATA', 'UPDATE', 'DELETE', and 'EXPORT CODE'. The table below shows 9 records of students. The bottom of the interface includes an 'Alert System Status: All Good' message, copyright information for MongoDB, Inc., and a chat icon.

	_id ObjectId	studentId String	name String	email String	course String	age Ir	
1	ObjectId('6a290017f8d7bd2...')	"S101"	"Ishal"	"ishal@gmail.com"	"BCA"	20	
2	ObjectId('6a290060f8d7bd2...')	"S102"	"Samuel"	"samuel@gmail.com"	"BTech"	22	
3	ObjectId('6a290132f8d7bd2...')	"S104"	"Rovin"	"rovin@gmail.com"	"MCA"	20	
4	ObjectId('6a290152f8d7bd2...')	"S105"	"Riya"	"riya@gmail.com"	"BCA"	21	
5	ObjectId('6a290176f8d7bd2...')	"S106"	"Sven"	"sven@gmail.com"	"BCom"	22	
6	ObjectId('6a2901a2f8d7bd2...')	"S107"	"Milin"	"milin@gmail.com"	"BTech"	20	
7	ObjectId('6a2901d2f8d7bd2...')	"S108"	"Ridha"	"ridha@gmail.com"	"MCA"	23	
8	ObjectId('6a2901f6f8d7bd2...')	"S109"	"Sona"	"sona@gmail.com"	"BCA"	22	
9	ObjectId('6a290226f8d7bd2...')	"S110"	"Sam"	"sam@gmail.com"	"BTech"	20	

One student record was deleted from the students collection. The collection was checked to ensure that the document had been removed successfully.